

Paul Chapron

Computer Science, Ph.D.
Social Sciences Modeling and Simulation

✉ paul.chapron@ign.fr
Univ Gustave Eiffel, Géodata Paris,
IGN, LASTIG, F-77454 Marne-la-Vallée, France
<https://www.umr-lastig.fr/paul-chapron/>

Current Position

since December 2017 Researcher at LASTIG laboratory

Special interest in : Social and spatial simulation models, model exploration via HPC methods, integration of model exploration into companion modelling, viability of ABM simulation models

Scientific Interests

Main Themes Agent Based Modeling and Simulation, Complex Systems, Social and Spatial Simulation, Simulation Models Evaluation and Validation, Companion Modelling.

Related Themes Sociology of Organizations, Quantitative Geography, Systems of Cities, Urban planning, Agricultural systems, Viability theory

Education and past Positions

February 2017 Postdoc at the Institute Complex Systems of Paris Ile de France (ISCPiF), CNRS UPS 3611. FUI Project Dynamicity. Calibration and robustness assessment of urban planning models. PI : Romain Reuillon.

Oct-Dec2016 **Postdoc**, *French National Institute of Geography and Forestry Information (IGN)*, Exploration of morphological diversity of buildings according to Right to Build regulation. Supervisor: Mickaël Brasebin and Julien Perret.

2015-2016 **Postdoc and Teaching Assistant**, *Institute of Geography and Durability, University of Lausanne*, Analysis and exploration of innovation geographical networks. Supervisor: Pr. Céline Rozenblat.

2013-2014 **Postdoc**, *Géographie-cités (UMR 8504 CNRS Paris I Paris VII)*, Modeling of system of cities evolution, within ERC grant project GeoDiverCity. Supervisor: Denise Pumain.

2007-2012 **PhD**, *Université Toulouse Capitole/IRIT UT1 UMR 5505, SMAC team*, Title: “Modélisation et analyse des organisations sociales: Propriétés Structurelles, régulation des comportements et évolutions”.Defended on May, 29th 2012.

2006-2007 **Research Master’s degree**, *Université Henri Poincaré, Nancy*.

2003-2006 **Computer Science Engineering degree**, *ESIAL (now Telecom Nancy), Nancy*.

Teaching

2018-2012 **Teacher (vacataire)**, ENSG, Université Gustave Eiffel.

Introduction au statistiques, encadrement et suivi de projet d'analyse spatiale.

Année Scolaire	Master 2 IGAST, ENSG, Université Gustave Eiffel	Mastère Spécialisé DE-SIGEO, ENSG, Université Gustave Eiffel	Total Heures
2018-2019	45h	9h	54h
2019-2020	42h	9h	51h
2020-2021	51h	12h	63h
2021-2022	50h	37h	87h

Softwares : R , QGIS

2015-2016 **Teaching Assistant**, University of Lausanne, IGD.

Involved in the **MOOC** "Analysis, Visualization and Simulation of Geographical Networks" of Pr. Céline Rozenblat. Video courses, courses materials and tutorials production.

Topics : Clustering techniques, Random network and ERGM models, Explanatory analysis of networks, Simple network dynamics simulation.

Softwares : TULIP (visualization), R (analysis and modeling), Netlogo (simulation)

2007-2012 **Teaching Assistant**, University of Toulouse Capitole.

Office software training	Bachelor students
Information Systems	Master students
Computer Architecture	Master students
Social Systems Modeling and Simulation	Master students

2006-2007 **Contractor Teaching Assistant**, Université de Nancy II.

Office software training	Bachelor students
--------------------------	-------------------

PhD supervision

2021-2024 **Jonathan Siliézar-Montoya**, UMRAE, Acoustics, Defended the 06/12/2024.

Multi-source modelling of urban sound environments

Internships supervision

2021 **Master Student**, Ludovic Moisan, Master 2.

Exploration des données issues de l'application NoiseCapture, par le prisme des sources sonores

2021 **Master Student**, Jonathan Siliézar-Montoya, Master 2.

Multi-sources modelling of urban acoustic landscapes(Projet LASSO, I-site FUTURE 2020)

2020 **Master Student**, Luke Riley, Master 2.

Joint characterization and visualization of urban climate and morphology

2018 **Master Student**, Vincent Delbar, Master 2.

Urban sprawl simulation scenarios in Occitanie 2040

Peer-reviewed Articles

International Journal Articles

- [1] Paul Chapron. Analyse de reseau de pouvoir au sein d'une organisation sociale. *Nouvelles Perspectives en Sciences Sociales*, 6(2):233–256, 2011.
- [2] Paul Chapron. Analyse structurelle des champs de capacité d'action dans les organisations sociales. outils analytiques pour la caractérisation de la coopération. *Studia Informatica Universalis*, 10(1):98–128, 2012.
- [3] Paul Chapron, Mathieu Leclaire, Arthur Scriban, Dominique Masse, Romain Reuillon, Abigail Fallot, and Etienne Delay. Fallow land simulations to fight against soil degradation: Engaging stakeholders with model exploration. *Journal of Artificial Societies and Social Simulation*, 29(1):3, 2026.
- [4] Clémentine Cottineau, Paul Chapron, and Romain Reuillon. Growing models from the bottom up. an evaluation-based incremental modelling method (ebimm) applied to the simulation of systems of cities. *Journal of Artificial Societies and Social Simulation*, 18(4):9, 2015.
- [5] Clémentine Cottineau, Romain Reuillon, Paul Chapron, Sébastien Rey-Coyrehourcq, and Denise Pumain. A modular modelling framework for hypotheses testing in the simulation of urbanisation. *Systems*, 3(4):348, 2015.
- [6] Christophe Sibertin-Blanc, Pascal Roggero, Françoise Adreit, Bertrand Baldet, Paul Chapron, Joseph El-Gemayel, Matthias Mailliard, and Sandra Sandri. Soclab: A framework for the modeling, simulation and analysis of power in social organizations. *Journal of Artificial Societies and Social Simulation*, 16(4):8, 2013.
- [7] Jonathan Siliézar, Pierre Aumond, Arnaud Can, Paul Chapron, and Matthieu Péroche. Case study on the audibility of siren-driven alert systems. *Noise Mapping*, 10(1), June 2023.
- [8] Kirana Widyastuti, Romain Reuillon, Paul Chapron, Wildan Abdussalam, Darmae Nasir, Mark E. Harrison, Helen Morrogh-Bernard, Muhammad Ali Imron, and Uta Berger. Assessing the impact of forest structure disturbances on the arboreal movement and energetics of orangutans—an agent-based modeling approach. *Frontiers in Ecology and Evolution*, Volume 10 - 2022, 2022.

French Journal Articles

- [9] Delay, Etienne, Chapron, Paul, Leclaire, Mathieu, Reuillon, Romain, and Faye, Ousmane. Entre terre et solidarité: la jachère collective comme lien social à dioline, sénégal. *Cah. Agric.*, 34:28, 2025.
- [10] Joseph El Gemayel, Paul Chapron, Françoise Adreit, and Christophe Sibertin-Blanc. Quand et comment les acteurs sociaux peuvent-ils coopérer ? Un algorithme de simulation pour la négociation de leurs comportement. *Revue d'Intelligence Artificielle*, 25, 2011.
- [11] Christophe Sibertin-Blanc, Françoise Adreit, Paul Chapron, Joseph El Gemayel, Matthias Mailliard, Pascal Roggero, and Claude Vautier. Compte-rendu d'une recherche interdisciplinaire entre sociologues et informaticiens : de la sociologie de l'action organisée au logiciel SocLab. *Technique et Science Informatiques*, 29(8-9):1081–1115, 2010.

International Conference Proceedings

- [12] Paul Chapron and Christophe Sibertin-Blanc. Analysis of the power network among the actors of a social organization (regular paper). In D. Kazakov and G. Tsoulas, editors, *AISB Symposium on Social Networks and Multiagent Systems, York, 04/04/2011-07/04/2011*, pages 2–7, <http://www.aisb.org.uk>, avril 2011. Society for the Study of Artificial Intelligence and the Simulation of Behaviour.

- [13] Joseph El Gemayel, Christophe Sibertin-Blanc, and Paul Chapron. Impact of tenacity upon the Behaviors of Social Actors (regular paper). In Alexis Drogoul, Benoit Gaudou, and Nicolas Marilleau, editors, *Applied Agent based simulator Engineering for Complex System study, Nagoya, Japan, 13/12/2009*, 2009.

Book Contributions

- [14] Françoise Adreit, Paul Chapron, Joseph El GEmayel, Pascal Roggero, Sandra Sandri, and Christophe Sibertin-Blanc. Une formalisation de la sociologie de l'action organisée : méta-modèle, simulation et étude analytique. In Denis Phan, editor, *Ontologies et modélisation par SMA en SHS (informatique et systèmes d'information, rta)*. Hermes Science Publications, 5 2014.
- [15] Clémentine Cottineau, Paul Chapron, Marion Le Texier, and Sébastien Rey-Coyrehourcq. Incremental Territorial Modelling. In *Geographical Modeling: Cities and territories*, number 1. Wiley, January 2020.
- [16] Clémentine Cottineau, Paul Chapron, Marion Le Texier, and Sébastien Rey-Coyrehourcq. Modélisation territoriale incrémentale. In Denise Pumain, editor, *La modélisation en géographie. Villes et territoires*, Méthodologies de modélisation en sciences sociales. ISTE, 2020.
- [17] Joseph El Gemayel, Christophe Sibertin-Blanc, and Paul Chapron. Impact of Tenacity upon the Behaviors of Social Actors. In Quan Bai and Noaki Fukuta, editors, *Advances in Practical Multi-Agent Systems*, volume 325, pages 287–306. Springer, <http://www.springerlink.com>, 2011.
- [18] Chapron Paul, Chérel Guillaume, Cottineau Clémentine, Cura Robin, Leclair Mathieu, Pumain Denise, Rey-Coyrehourcq Sébastien, Reuillon Romain, Schmitt Clara, and Swerts Elfie. *Urban Dynamics and Simulation Models*. Springer International Publishing, 2017.

Other Publications

Workshop and Communications

- [19] Françoise Adreit, Paul Chapron, , Christophe Sibertin-Blanc, Pascal Roggero, and Sandra Sandri. Une ontologie des systèmes d'actions concrets. Rencontres interdisciplinaires sur les systèmes complexes naturels et artificiels - Journées de Rochebrune, Rocchebrune, 19/01/09-23/01/09., 2008.
- [20] Françoise Adreit, Paul Chapron, Pascal Roggero, and Christophe Sibertin-Blanc. Régulation des organisations et rationalité des acteurs. XVIII Congrès de l'AISLF, Istanbul (conférence sans actes), 07/07/08-11/07/08, 2008.
- [21] Paul Chapron. Analysing the dynamics of a social organization by means of its transfer function. International Conference "Interactions in Complex Systems" and 12th International Workshop "Spatial Econometrics and Statistics", June 17-19, Université d'Orléans <http://www.univ-orleans.fr/mapmo/peps/?q=content/international-conference-interactions-complex-systems-and-12th-international-workshop> http://www.univ-orleans.fr/mapmo/peps/sites/default/files/conferences/programme_ISC_Print.pdf, 2013.
- [22] Paul Chapron. Diversité des formes d'îlots urbains à partir d'un Plan Local d'Urbannisme. Journée de la recherche IGN, 23 mars 2017. <http://recherche.ign.fr/jr/jr17.php>, 2013.
- [23] Paul Chapron and Joseph El Gemayel. Social Actors Rationality within Organizations. Simulation of Complex Social Systems, (conférence sans actes) Sevilla, Spain, 09/11/2009, 2009.
- [24] Clémentine Cottineau and Paul Chapron. Back in the USSR: a series of models for simulating urban evolution. Urban Systems Modeling: advances from GeoDiverCity and MECHANICITY

- ERC programmes. Satellite meeting of the European Conference on Complex Systems 2013, Barcelona (Spain), 2013.
- [25] Clémentine Cottineau and Paul Chapron. Step-wise modeling. An Incremental method for modeling cities trajectories in the post-Soviet urban system. 18th European Colloquium in Theoretical and Quantitative Geography (ECTQG) , 2013.
- [26] Clémentine Cottineau and Paul Chapron. Evaluation & calibration for the comparison of abms of cities' trajectories. Paper Session: 2539 Geosimulation Models 3: Applications - Macro,AAG Meeting 2014, Tampa (Florida) <http://meridian.aag.org/callforpapers/program/SessionDetail.cfm?SessionID=19892>, 2014.